ETH Zürich
Laboratory of Ion Beam Physics

Integrated Computational Infrastructure and Hardware Control

Semester Project

Author
Chanel Dobler

Supervisor
Dr. Lukas Wacker

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1 SMC EX260-SDN1 DeviceNet

1.1 Physical Switch Settings

1.1.1 Communication Speed

The communication speed is set with the physical switches No. 3 and 4. The default value is 125kbps. [1, page 16]

Bitrate	Switch No.	
kbps	3	4
125	0	0
250	0	1
500	1	0

Table 1: Bitrates from Switch Settings

1.1.2 DeviceNet Address

The DeviceNet address is set with the physical switches No. 5 to 10. The default address is 63. [1, page 16]

Address	Switch No.					
-	5	6	7	8	9	10
0	0	0	0	0	0	0
1	0	0	0	0	0	1
2	0	0	0	0	1	0
...				...		
62	1	1	1	1	1	0
63	1	1	1	1	1	1

Table 2: Addresses from Switch Settings

1.2 Messaging Protocol

A message consists of a connection ID and a data field. Wait $\geq 1s$ for a response when you send a message. [2, chapter 2, page 11]

1.2.1 Connection ID (CID)

The CID counts 11 bits. The structure depends on the message group. Only group 2 is relevant for this project. The CID for group 2 contains the group ID (2 bits), the MAC ID (6 bits) and the message ID (3 bits). [2, chapter 2, pages 6-7]

	Group ID		MAC ID						Message ID		
Bit No.	10	9	8	7	6	5	4	3	2	1	0

Table 3: Group 2 CID Structure

Message ID	Function
4	Master's Explicit Request
6	Master/Slave Connection Management
7	Duplicate MAC ID Check

Table 4: Relevant Group 2 Message IDs

1.2.2 Data Field

The data field may include some flags. The R flag indicates a request or response. The F flag indicates a complete or fragmented message. The X flag is a transaction ID that's echoed in a response to match a request. [2, chapter 2, pages 22-24]

Flag	0	1
R	Request	Response
F	Complete	Fragmented
X	-	-

Table 5: Data Field Flags

1.3 Communication Process

The communication process involves checking for a unique MAC ID before allocating a master/slave connection. The connection times out after a few seconds. If no response is received for a master's explicit request, the master/slave connection requires reallocation.

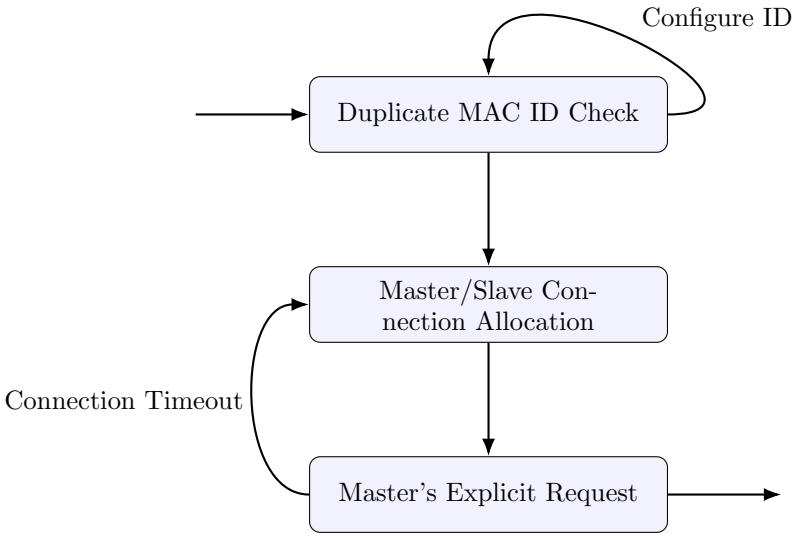


Figure 1: Communication Setup Process

1.3.1 Duplicate MAC ID Check

The message ID 7 is reserved for the duplicate MAC ID check in group 2. It requires human intervention to configure a unique MAC ID as it's inevitable that two modules on the same link will be assigned the same address. The duplicate MAC ID check has to be transmitted two consecutive times without receiving a subsequent request or response message before establishing a connection. [2, chapter 2, pages 11-14]

Group ID		MAC ID						Message ID		
1	0	0	0	0	0	0	0	1	1	1

Table 6: CID for Duplicate MAC ID Check

Byte No.		
0	R	Physical Port Number
1		Vendor ID
2		
3		Serial Number
4		
5		
6		

Table 7: Data Field for Duplicate MAC ID Check

1.3.2 Master/Slave Connection Management

The message ID 6 is reserved for the master/slave connection management in group 2. The service code 0x4B indicates the allocation of a master/slave connection and the service code 0x4C indicates the release. The class ID 3 directs the message at the DeviceNet object. There exists exactly one DeviceNet object per physical attachement to the DeviceNet. Therefore, the instance ID is always 1. The allocation choice byte denotes an explicit message and I/O connections from the predefined master/slave connection set. The acknowledge suppression bit has to be 0 for the release. [2, chapter 3, pages 13-24]

Group ID		DeviceNet Address						Message ID		
1	0	1	1	1	1	1	1	1	1	0

Table 8: CID for Master/Slave Connection Management

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4B]	
2	Class ID [0x03]		
3	Instance ID [0x01]		
4	Allocation Choice		
5	DeviceNet Address		

Table 9: Data Field for Master/Slave Connection Allocation

Byte No.		
0	F	X MAC ID
1	R	Service Code [0x4C]
2		Class ID [0x03]
3		Instance ID [0x01]
4		Release Choice

Table 10: Data Field for Master/Slave Connection Release

Bit No.	Connection
0	Explicit Message
1	Polled
2	Bit Strobed
3	Multicast Polling
4	Change of State
5	Cyclic
6	Acknowledge Suppression
7	Reserved

Table 11: Allocation/Release Choices Byte

1.3.3 Master's Explicit Request

The message ID 4 is reserved for the master's explicit request in group 2. The service code 0x10 indicates a write to a single attribute and the code 0x0E a read. The class ID 9 represents the discrete output points. The class ID 4 represents the Assembly Object. The instance ID 37 corresponds to the 32 solenoid output points. The value bytes act as a mask to set all 32 channels at once. The instance ID corresponds to the desired channel. [2, chapter 2, pages 36-37] [1, pages 20-22]

Group 2		DeviceNet Address						Message 4		
1	0	1	1	1	1	1	1	0	1	0

Table 12: CID for Master's Explicit Request

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		

Table 13: Data Field for Master's Explicit Read

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		
5	Value		

Table 14: Data Field for Master's Explicit Write

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		

Table 15: Data Field for Master's Explicit Read All

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		
5	Value		
6			
7			
8			

Table 16: Data Field for Master's Explicit Write All

2 Keller PAA-4LD I^{2}C

2.1 Request Measurements

The command to request a measurement is 0xAC. Wait $\geq 8ms$ before reading the measurement. [3, pages 8-9]

2.2 Read Measurements

The response contains the status (1 byte) and the measurement (2 bytes). The status carries a checksum error indication at bit 2. The two data bytes are stitched together and then scales with $(p - 16384) * (p_{\max} - p_{\min})/32768 + p_{\min}$ for the pressure. The absolute pressure ranges from $p_{\min} = 0$ bar to $p_{\max} = 3$ bar. [3, pages 8-9]

3 Autonics TM4-N2SB RS485

Code	Address (Channel i)	Description (Run=0, Stop=1)
1	0x0002 * i	Read Heating Status
5	0x0002 * i	Write Heating Status
3	0x03E8 * i	Read Set Temperature
6	0x03E8 * i	Preset Temperature
4	0x03E8 + 6 * i	Read Temperature

Table 17: Function Codes and Addresses [4]

4 Elemental Analyzer (EA) TCP/IP

The EA device communicates over the TCP/IP protocol. The messages are sent to the IP address of the EA device via a dedicated port.

Message	Description
STRT	Start device
SLEEPON	Send device to sleep
?STS	Get device status
SEQON	Start processing
SEQOFF	Halt processing
SEQ	Get process status
?SINX	Get sample index `i`
?NAM `i`	Get sample name
?WGH `i`	Get sample weight
?PCT `i` C	Get sample C%
?PCT `i` N	Get sample N%

Table 18: Messages for the EA Device

5 Graphitization

The graphitization process is split into 3 stages: Conditioning the device, loading the samples and the reaction. Each stage is a wrapper for their procedures. Some procedures can be executed in parallel because they operate on separate sections of the graphitization device.

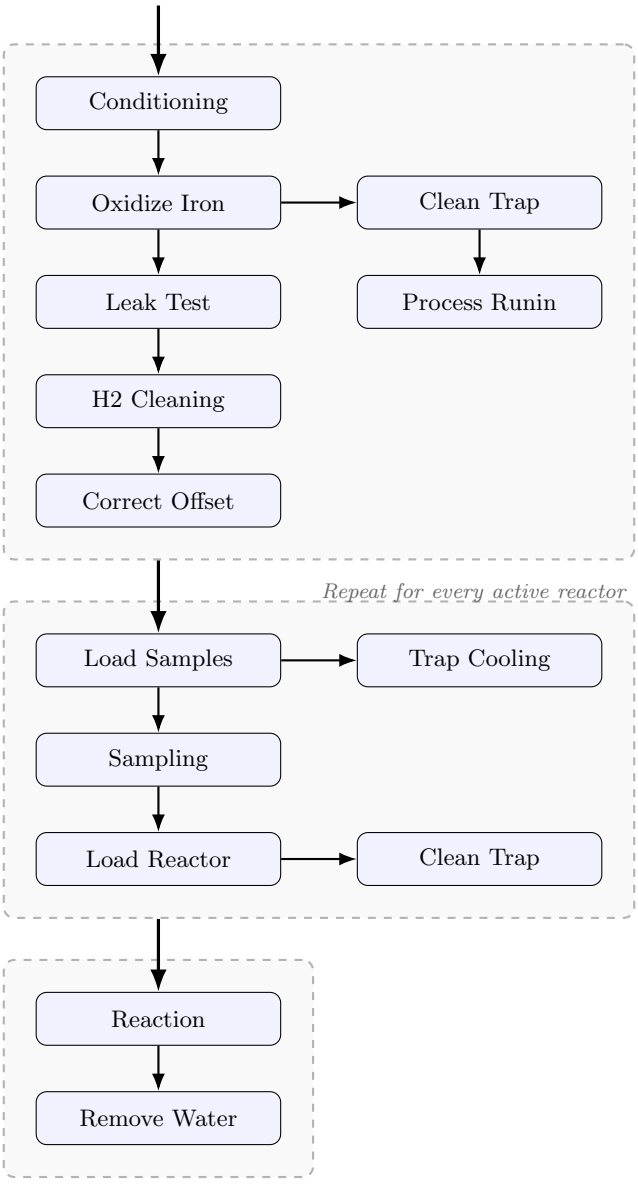


Figure 2: Graphitization Process Overview

5.1 Conditioning

5.1.1 Oxidize Iron

1. Open the line valves
2. Set the ovens to 500°C
3. Run the ovens
4. Wait for the ovens to reach 500°C
5. Lift the ovens to the reactors
6. Wait 5min for the irons to preheat
7. Queue Clean Trap and Process RUNIN

5.1.2 Leak Test

8. Open the reactor valves and the vacuum valve
9. Wait 3min for the reactors to be evacuated
10. Close the reactor valves and the vacuum valve
11. Check the pressure difference over 5min

5.1.3 H2 Cleaning

Repeat H2 Cleaning (12. - 18.) 3 times

12. Open all active reactor valves
13. Open the hydrogen valve
14. Wait until one of the reactors reaches 800mbar
15. Close the hydrogen valve
16. Wait 5min
17. Open the vacuum valve
18. Wait 1min for the reactors to be evacuated
19. Close the vacuum valve
20. Close all active reactor valves

5.1.4 Correct Offset

21. Reset the pressure offsets
22. Open the reactor valves and the vacuum valve
23. Wait 20s for the reactors to be evacuated
24. Store the remaining pressure as offset
25. Close the reactor valves and the vacuum valve

5.2 Load Samples

Repeat Load Samples (24. - 59.) for every active reactor

26. Wait for the EA status "STS READY"
27. Store the sample index, name and weight
28. Set Trap Cooling active
29. Queue Trap Cooling

5.2.1 Sampling

30. Wait for the trap to reach 50°C
31. Start the EA
32. Wait for the EA status "RUN"
33. Start processing the sample
34. Wait for the processing status "10"
35. Open the sampling valve
36. Wait for the processing status "15"
37. Close the sampling valve
38. Stop processing the sample
39. Store the sample C% and N%
40. Set Trap Cooling inactive
41. If the sample name starts with "first" (any capitalization), go back to Load Samples (24. - 59.)
42. If the sample name starts with "pre" (any capitalization), queue Clean Trap and go back to Load Samples (24. - 59.)

5.2.2 Load Reactor

43. Open the trap valve and the vacuum valve
44. Wait 3s for the trap to evacuate
45. Open the reactor valve
46. Wait 10s for the reactor to evacuate
47. Close the vacuum valve
48. Set the trap to 420°C
49. Start heating the trap
50. Wait for the trap to reach 420°C
51. Wait 10s for the sample to be loaded to the reactor
52. Close the reactor valve and trap valve
53. Stop heating the trap
54. Queue Clean Trap
55. Store the pressure as CO2 pressure
56. Open and close the vacuum valve and the line valve of the current reactor alternately until the pressure reaches 570mbar
57. Store the pressure as CO2 pressure final
58. Open the reactor valve and the hydrogen valve
59. Wait until the pressure reaches 3.3 * CO2 pressure final - 200mbar
60. Open and close the vacuum valve and the line valve of the current reactor alternately until the pressure reaches 3.3 * CO2 pressure final
61. Store the pressure as H2 pressure

5.3 Reaction

5.3.1 Reaction

62. Start the peltier coolers
63. Set the oven temperature to 580°C
64. Wait for the ovens to reach 580°C
65. Wait 2h for the reaction
66. Lower the ovens from the reactors
67. Stop the ovens

5.3.2 Remove Water

68. Open the reactor valves and the vacuum valve
69. Wait 7mins for the reactors to be evacuated
70. Close the reactor valves, the line valves the vacuum valve
71. Stop the peltier coolers

5.4 Separate Procedures

5.4.1 Clean Trap

1. Set the trap to 450°C
2. Start heating the trap
3. Open the sampling valve for helium from the EA
4. Wait for the trap to reach 450°C
5. Wait 1min for the trap to be flushed
6. Stop heating the trap
7. Wait 20s
8. Close the sampling valve

5.4.2 Trap Cooling

Repeat (1.) while trap cooling is active

1. If the trap is above 10°C open the vortex valve else close the vortex valve
2. Close the vortex valve

5.4.3 Process RUNIN

Repeat Process RUNIN (1. - 3.) for every active reactor

1. Wait for the EA status "STS READY"
2. Store the sample index and name
3. If the sample name starts with "runin" (any capitalization) start the EA

6 Graphical User Interface (GUI)

The GUI is built from the PySide 6 library.

6.1 Control

The control class is central to the GUI. It holds several other classes such as the graphitization procedures, the samples table, the pressure plot and the device schematic.

6.1.1 Schematic

The schematic represents the graphitization device. It shows the reactors, heaters, coolers and other connections such as the vacuum device, H_2 supply and the EA. The reactors display the current pressure if they are active. They can be deactivated with a click on the button "deactivate" and reactivated at a later time. If a line of reactors is deactivated, that side of the heaters and coolers is automatically deactivated during the procedures. The heaters and coolers display if they are running in the grey field. The valves can be manually opened or closed at any time with a click.

6.1.2 Graph

The graph plots the pressure history of the active reactors in different colors. It's updated with every pressure measurement poll.

6.1.3 Table

The table is populated during the graphitization and holds the sample data such as the name, weight, pressure, CO_2 and H_2 concentrations.

6.1.4 Buttons

The GUI holds 3 buttons: Start, stop and reset. Clicking start runs all stages of the graphitization. The stop button can be clicked at any time to stop the process, turn off the heaters and coolers and close all valves. With the reset button, the graph, the status displays and the table are cleared and the pressure offset is reset.

6.1.5 Displays

There are 4 displays: Conditioning, Load Samples, Reaction and Separate. The headers of Conditioning, Load samples and Reaction can be clicked to run the corresponding stage. The status is displayed underneath the headers. The status shows what process was started, if the process was stopped or if the stage is completed.



Figure 3: Control Window

7 Threading

The GUI runs on the main thread. Further threads are introduced to keep the GUI responsive at all times. To avoid corrupting messages to the hardware, every device has its own lock. Only one thread at a time can acquire the lock and communicate to the device. There are dedicated threads for every device that regularly poll new data. The procedures are executed in a separate thread that can request writes to the devices. The procedures emit a signal to update the status in the GUI. An additional thread runs the separate processes in parallel when called from the procedures.

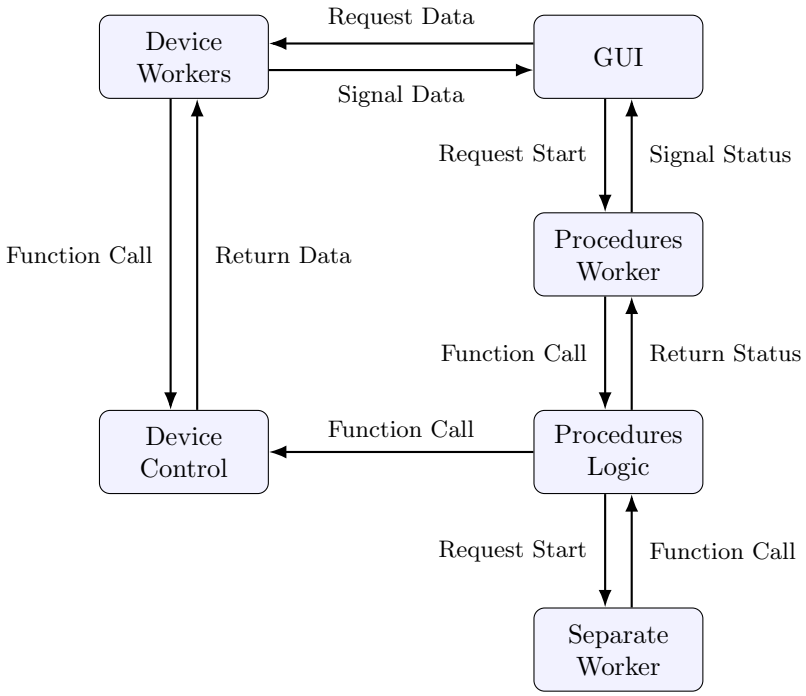


Figure 4: Thread Interactions

8 Configurations

8.1 SMC

DeviceNet Address	MAC ID	Bitrate
63	0	125000

Table 19: DeviceNet Configuration

Name	ID
Reactor 1	1
Reactor 2	2
Reactor 3	3
Reactor 4	4
Reactor 5	5
Reactor 6	15
Reactor 7	14
Reactor 8	13
Reactor 9	12
Reactor 10	11
Line 1	6
Line 2	10
Vacuum	7
Trap	8
H2	9
He	16
-	17
Vortex	18
Lift	19

Table 20: Valve Layout

8.2 Keller

Reactor No.	Address
Reactor 1	0x50
Reactor 2	0x51
Reactor 3	0x52
Reactor 4	0x53
Reactor 5	0x54
Reactor 6	0x55
Reactor 7	0x56
Reactor 8	0x57
Reactor 9	0x58
Reactor 10	0x59

Table 21: Reactor Addresses

8.3 Autonics

Port	Address	Baudrate
/dev/ttySC0	1	9600

Table 22: Instrument Configuration

Name	ID
Trap	1
Oven 1	2
Oven 2	3

Table 23: Channel Configuration

9 Future Work

9.1 Prompts

At different stages, the graphitization may stop due to some conditions not being met. Instead, a prompt should ask the user if they want to retry, continue or cancel.

9.2 Configurations Tab

A separate configurations tab should display the parameter settings such as durations, temperatures, etc. When hovering over these parameters, a description should appear. The parameter settings should be editable and will be saved to the .json file.

9.3 Storing Data

The sample data and the completed stages should be saved to a file and loaded in case of an unexpected failure.

References

[1] SMC Corporation, *SI Unit for DeviceNet Operation Manual: EX260 Series*, Jul. 2016, available at https://www.smcworld.com.

[2] ODVA, *The CIP Networks Library, Volume 3: DeviceNet Adaptation of CIP*, 1st ed., Nov. 2018, specification document.

[3] KELLER AG für Druckmesstechnik, *Description of the Communication Protocol: Series 4 LD...9LD OEM Pressure Transmitter*, 2nd ed., Aug. 2017. [Online]. Available: Available at https://keller-pressure.com/

[4] Autonics Corporation, *User Manual for Communication: Multi-CH Modular Type Temperature Controller TM Series*, 1st ed., May 2020, available at https://www.autonics.com.

Generative AI, specifically Gemini, was used for consultation. The graphitization process is based on legacy LabVIEW code.

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Eidgenössische Technische Hochschule Zürich
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